picBunch: a free-space FFT convolution for the initial field of a uniformly moving particle bunch, and its measured cost and accuracy

picBunch authors

Run `pr24-20261102`

1 Introduction

When a particle-in-cell (PIC) simulation starts, the bunch of particles must carry an initial electromagnetic field, or the first push already acts on the wrong force. The field of a moving bunch in free space is known in closed form for each point charge, but a simulation reads that field on a staggered mesh with thousands of observation points, so it is computed by a fast sum over the source charges rather than by evaluating the closed form everywhere. We do that sum as a TSC deposit onto a zero-padded grid, a sampled exact TSC-convolved kernel, and a free-space FFT convolution, with staggered Yee read-out. This paper gives the exact discrete recipe a re-implementer needs, and it measures the cost and the accuracy of that recipe on a single V100, naming the reference for every error statement.

The established workaround avoids computing the initial field altogether: one starts from a field-free state and marches the bunch forward, letting the fields build up in the first few steps. The present-time field of a uniformly moving bunch has no transient, so this history has no physical basis. We measure how well the discrete method reproduces that field and how much it costs, and we state the conditions under which it fails (Section 5).

2 Field definition

The target quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p, and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation point $\mathbf{x}_{\text{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\text{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\text{point},c}\big(\mathbf{x}_{\text{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 3) and K_{point} is the per-unit-charge field of one uniformly moving charge. Because the integral is linear in the source charge, the sum over particles is exact; the whole design is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge [extraction worker, 2026]. For constant velocity the field points to the instantaneous position. In the lab frame, with $\mathbf{R} = \mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \text{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2\, \text{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \text{invg}^2\, \mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class; the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The reference normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta} \times E_n$ [extraction worker, 2026]. The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m and $c = 299792458$ m/s. The factor h^3 (Section 3) is missing from the original delivered mesh and is the single most important porting fix [extraction worker, 2026, worker, 2026a].

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis [extraction worker, 2026]:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Method

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, and a free-space FFT convolution. Let N be the bunch-box nodes per axis, $M = \mathrm{pad} \cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at $ih\,\mathbf{cell}$), and $\mathbf{cell} = (dx, dy, dz)$. All precision-critical work is in double [worker, 2026b, extraction worker, 2026].

3.1 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h [extraction worker, 2026]. The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\mathrm{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd` [extraction worker, 2026].

3.2 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\,\mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.4). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide [worker, 2026c]. Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT [extraction worker, 2026].

3.3 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution. The vendor plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transformations [extraction worker, 2026]. The unnormalized round trip is scaled by M^3; the code divides by h^3M^3 in the read-out [extraction worker, 2026, worker, 2026a].

Padding. The grid must be zero-padded to pad ≥ 2. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box [worker, 2026d]. At pad $= 2$ the result agrees with pad $= 3$ to 8.4×10^{-16} and with a direct non-cyclic sum to 2.8×10^{-17} [worker, 2026d,b]. At pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$) [worker, 2026d]. The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error [worker, 2026d].

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real [worker, 2026d]. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components [worker, 2026d].

The h^3 factor. The delivered mesh applied only the $1/M^3$ scale, which equals the correct value only at $h = 1$. For general h the field must be multiplied by h^3 [extraction worker, 2026]. Missing it leaves the field $1/h^3$ too large: at $h = 1/2$ by a factor of 8, at $h = 1/4$ by a factor of 64 [extraction worker, 2026, worker, 2026a]. Section 6 gives the measured per-h series and the refinement gate [worker, 2026a].

3.4 Read-out

The six field components live at the staggered Yee positions, in cell units [extraction worker, 2026]:

$$\begin{array}{llllll} E_x & (0.5,0,0) & E_y & (0,0.5,0) & E_z & (0,0,0.5) \\ B_x & (0,0.5,0.5) & B_y & (0.5,0,0.5) & B_z & (0.5,0.5,0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed [extraction worker, 2026].

3.5 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units [worker, 2026c]. At a fixed distance in cells the error is therefore independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1, 1/2, 1/4, 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4 [worker, 2026c]. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The only knob is r/h: reaching 10^{-3} perpendicular at $\gamma = 10$ for $r = 8$ cells needs $h \approx 1/29$, 2.4×10^4 times the M^3 nodes of the $h = 1$ run [worker, 2026c, Orchestrator, 2026a].

4 Measured scaling

All timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0 [worker, 2026b]. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double [worker,

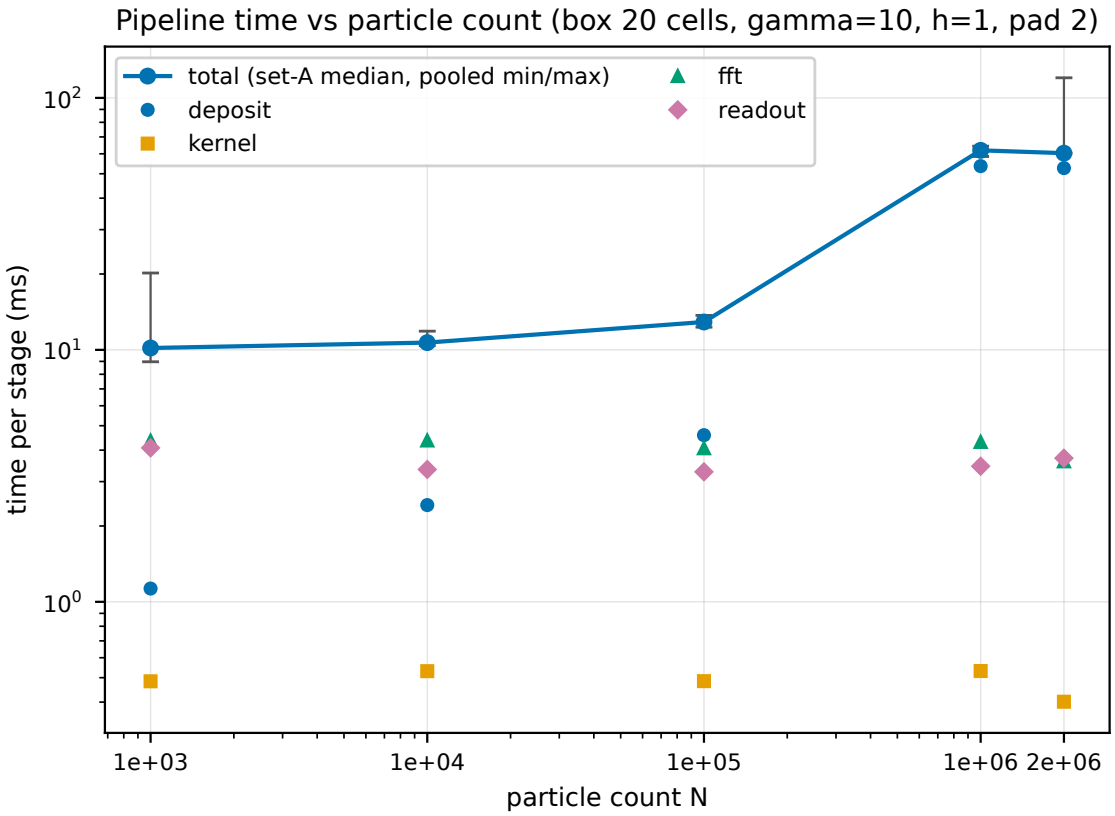


Figure 1: Pipeline time vs particle count. `rosi_v100`, GPU 0 (V100-SXM2-32GB, idle), CUDA 12.6, Release, $N \in \{10^3, 10^4, 10^5, 10^6, 2\cdot10^6\}$, source box 20 cells, $\gamma = 10$, $h = 1$, padding 2, double. Blue is the total time; grey bars span the pooled min/max over the four $N = 2\cdot10^6$ repeats. The other series are the per-stage medians, side by side, not stacked. Log-log [worker, 2026b].

2026b]. Each configuration ran three timed repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max [worker, 2026b]. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run [worker, 2026b]. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 2, a different measurement path.

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold [worker, 2026b,e]. This confirms the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff.

The scaling is in two regimes, with the crossover set by the box, not the particle count [worker, 2026b]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms) [worker, 2026b]. At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× because the deposit stays constant [worker, 2026b].

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB [worker, 2026b]. Memory scales with the padded bunch box, not the global field.

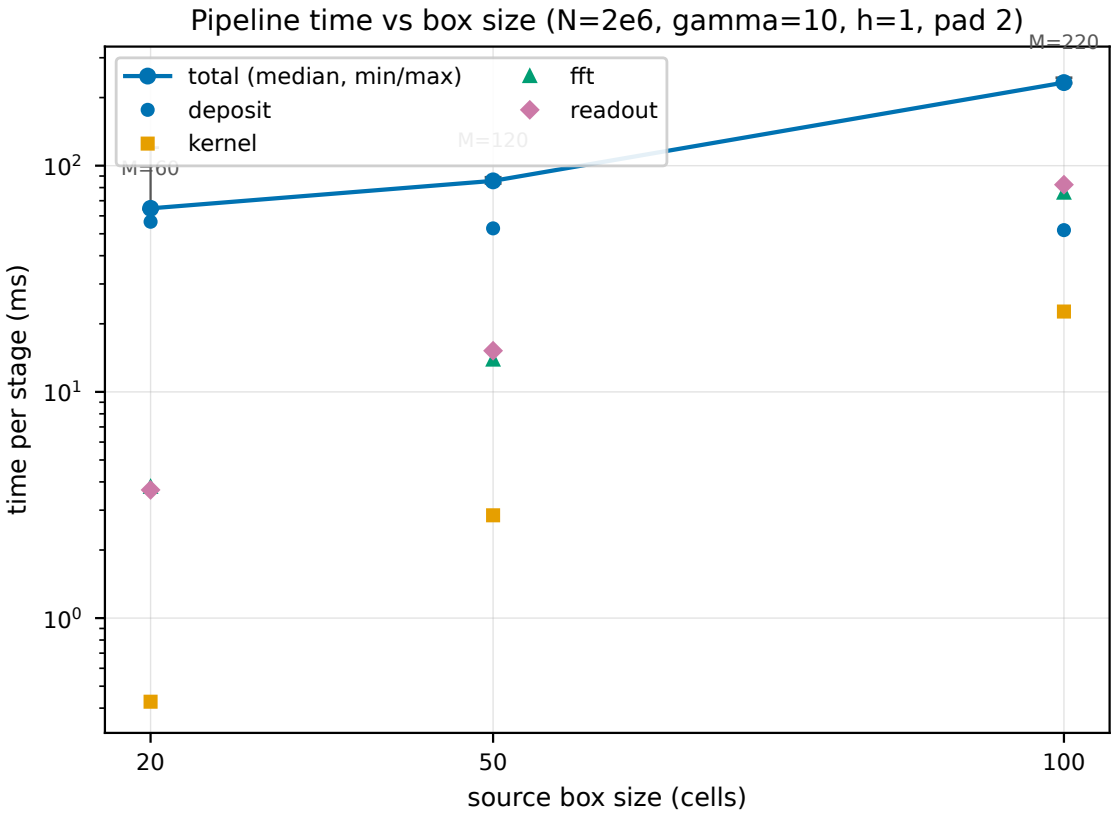


Figure 2: Pipeline time vs source-box size. `rosi_v100`, GPU 0, CUDA 12.6, Release, $N = 2 \cdot 10^6$, $\gamma = 10$, $h = 1$, padding 2, double. Box sizes 20/50/100 cells with padded FFT array $M = 60/120/220$ per axis. Blue is the total; the other series are the per-stage medians [worker, 2026b].

Table 2 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 $\rightarrow$ 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 1; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer [worker, 2026f,b].

5 Limitations

The mesh is a far-field method. Its limits are set by r/h and γ, and they are measured, not assumed. The near shell is not implemented in v0.1: the delivered near-field tables were off by 1.9×10^{-2} to 3.3×10^{-1} against the 3×10^{-4} target on all six [Orchestrator, 2026b]. The perpendicular direction at $\gamma = 10$ does not converge at the natural cell size: 1.5×10^{-2} at $d = 48$, aspect 1, against the point field, of which the TSC smoothing alone is 1.48×10^{-2} (Table 3) [worker, 2026c, Orchestrator, 2026c]. At $\gamma = 100$ the perpendicular error is 3.75×10^{-1} at $d = 48$ and does not converge as d grows [worker, 2026c, Orchestrator, 2026d]. At aspect 4, $\gamma = 1$ already fails the gate at $d = 48$ (2.4×10^{-3}) [worker, 2026c]. The detailed discussion is in Section 8.

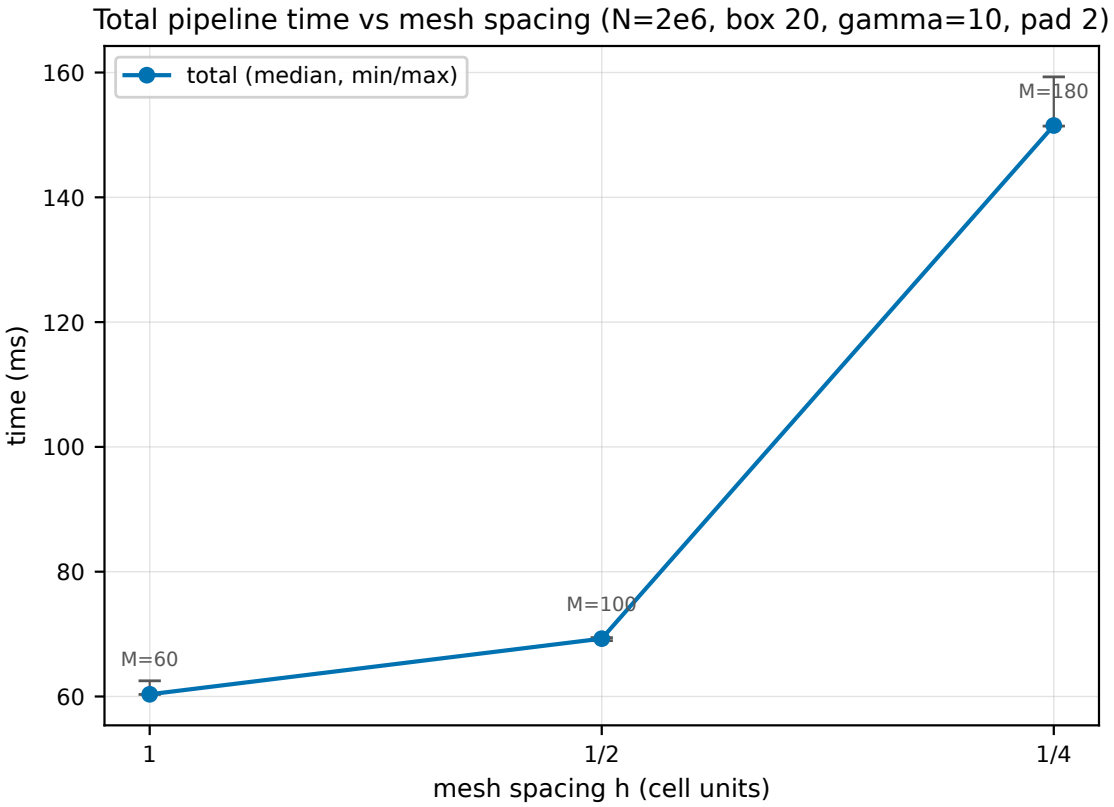


Figure 3: Total pipeline time vs mesh spacing h. `rosi_v100`, GPU 0, CUDA 12.6, Release, $N = 2 \cdot 10^6$, source box 20 cells, $\gamma = 10$, padding 2, double, $h \in \{1, 1/2, 1/4\}$, padded array $M = 60/100/180$. Blue is the total (median, min/max bars) [worker, 2026b].

6 Accuracy

Each error figure names its reference (M mesh, T continuum, P point). The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

6.1 The reference and the two metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map [pr24w53 ref, 2026]. Its quadrature error is below the 3×10^{-4} gate in every configuration measured here [worker, 2026c].

Two metrics are reported and are not interchangeable. The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, |\text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the $+1$ denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate [worker, 2026f]. The relative per-component error $|\text{ref} - \text{mesh}|/\max(|\text{ref}|, |\text{mesh}|)$, over components where $|\text{ref}| \geq 10^{-3}$ of the local field, is the meaningful one. This section reports relative error as the metric of record.

6.2 Reference-separated error at $d = 48$

Table 3 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M - P = 1.5 \times 10^{-2}$) is about 58× the mesh against its own target ($M - T = 2.6 \times 10^{-4}$, an L2 ratio); the

Table 1: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0 [worker, 2026b]. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

Table 2: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, shipped CLI `-tolerance` 10^{-3} [worker, 2026f]. This is a different measurement path from the compute-only Table 1: the totals here include the writer (the dominant added term), and the per-stage split is deposit/kernel/FFT/readout/writer. The $N = 10^6$ total is 47.15 ms here versus 61.92 ms in the compute-only campaign, which excludes the writer and uses a different timing boundary.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

difference is the TSC smoothing, which $M{-}P$ inherits and $M{-}T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M{-}T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ [worker, 2026c].

6.3 The independent validation

A second, independent pass measured the shipped CLI end to end: `picBunch` at `bfaa1e26` (field path unchanged through v0.1.3), rosi_v100, `indfield.py` order 12 refine 3, $d = 2$ to 32 cells, $\gamma \in \{1, 10\}$, aspect 1 and 4, single particle and three-particle cloud [worker, 2026f].

Table 4 confirms the reference-separated picture independently. $\gamma = 1$ crosses 3×10^{-4} at $d = 12$ (aspect 1, 1.1×10^{-4}) and $d = 32$ (aspect 4, 2.6×10^{-4}); at $\gamma = 10$ the relative error floors near 6×10^{-3} at $d = 32$ for all four cases and does not cross 3×10^{-4} within the swept range. The near shell is far worse, with relative error $O(0.4)$–0.8 at $d = 2$. The window starts at $d = 2$, so the TSC support interior ($|s| < 1.5$ cells) is not covered [worker, 2026f].

The point-field column (Table 3, $M{-}P$) is the mesh against the un-smoothed field. It carries the TSC smoothing in addition to the deposit error, and the two never agree for $\gamma h \gg 1$. The first half's distance sweep (Fig. 4) plots this point-field comparison; the $M{-}T$ column of Table 3 is the mesh against its own target.

6.4 Crossover radii

At $\gamma = 1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d = 8$; the two are interchangeable there. From $\gamma = 10$ up, against the point field the perpendicular direction does

Table 3: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; the single-component worst for $\gamma = 1$, aspect 4, $M-T$ is 2.65×10^{-5} and for $\gamma = 10$, aspect 1 is 1.4×10^{-3}) [worker, 2026c]. $M-T$: mesh vs TSC-convolved continuum (the target). $M-P$: mesh vs point field. $T-P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1, the gap between $M-P$ and $M-T$ is the smoothing (1.5×10^{-2} vs 2.6×10^{-4}, an L2 ratio of 58×), not the mesh.

γ	aspect	$M-T$	$M-P$	$T-P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

Table 4: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the shipped CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; rosi_v100, `indfield.py` order 12 refine 3 [worker, 2026f]. "s/c" = single / cloud. $\gamma = 10$ does not cross 3×10^{-4} within 32 cells; $\gamma = 1$ crosses at $d = 12$ (aspect 1) and $d = 32$ (aspect 4).

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

not reach 3×10^{-4} within 48 cells; against the target the perpendicular crossover is 48 ($\gamma = 10$, aspect 1) or not reached. The axial direction stays at 6 against the target at every γ [worker, 2026c].

6.5 The disagreement is per-source

On a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$), the disagreement with the exact-quadrature reference does not grow with particle count: outside the TSC support the worst per-component error stays in the 10^{-3} to 3×10^{-2} band across $N = 1$ to 1000 [worker, 2026g]. The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.42–0.65 cells from a source, inside that source's TSC support. A flat per-source floor is what a kernel or reference-type difference produces; a charge-weighting defect would grow with N. The measured disagreement is the in-support, near-source regime, and it does not depend on the number of sources [worker, 2026g].

6.6 The h^3 defect and its correction

Section 3 derives the h^3 factor and the $1/h^3$ overshoot of the delivered mesh. Fig. 5 is the measured per-h series, worst per-component relative error against `indfield.py`, at fixed physical separation, $\gamma \in \{1, 10\}$, aspect 1 and 4, rosi_v100. The per-h runs are at fixed physical separation, so they are not a scale-invariance demonstration (Section 3.5); they show that the after-fix field is stable to about 9×10^{-4} over $h = 1$ to 1/2 (the refine gate, below), while the pre-fix field, $1/h^3$ too large in absolute terms, has a relative error that rises toward 1 as h is refined (only the absolute error scales as h^{-3}; the relative error saturates because the reference is not scaled with it) [worker, 2026a].

The correction is verified three ways [worker, 2026a]. The refinement gate $|\mathbf{h} = 1 - \mathbf{h} =$

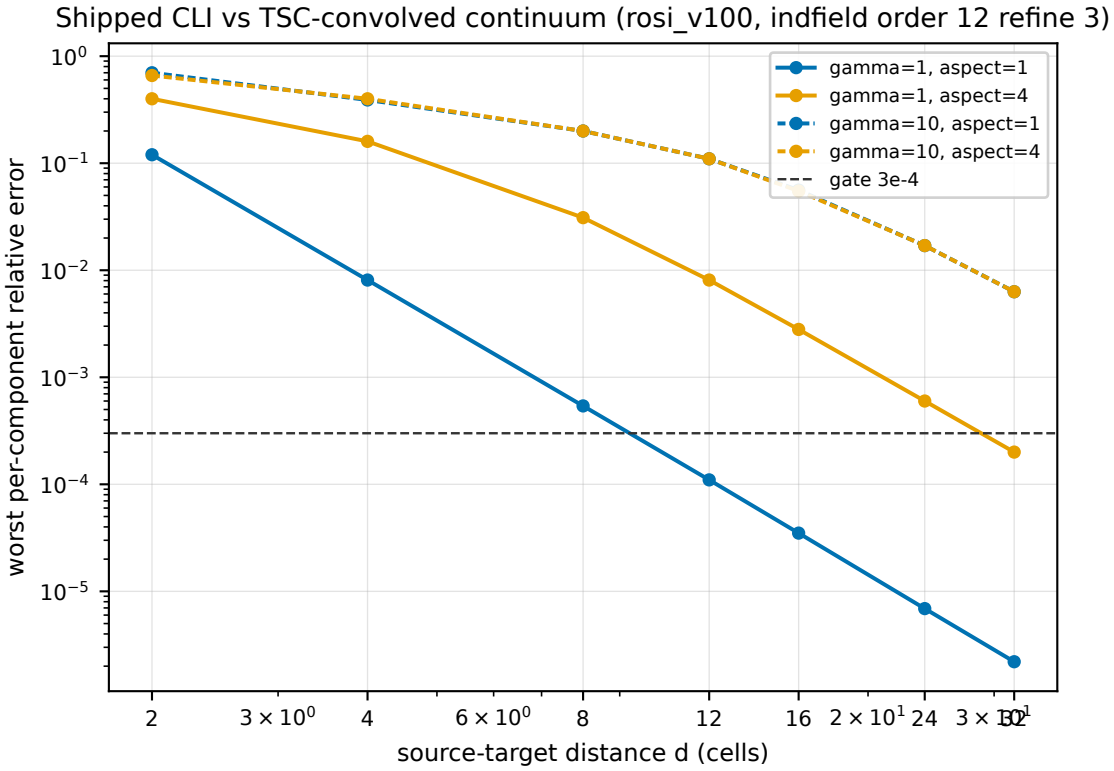


Figure 4: Worst per-component relative error of the shipped CLI against the TSC-convolved continuum, per d in cells, single-particle series only (the three-particle cloud is in Table 4); rosi_v100, `indfield.py` order 12 refine 3 [worker, 2026f]. Solid, $\gamma = 1$; dashed, $\gamma = 10$; blue, aspect 1; orange, aspect 4. $\gamma = 1$ crosses the $3{\cdot}10^{-4}$ gate at $d = 12$ (aspect 1) and $d = 32$ (aspect 4); $\gamma = 10$ floors near $6{\cdot}10^{-3}$ at $d = 32$ and never crosses within the swept range.

$1/2|/(1 + \max)$ is 9.13×10^{-4} after the fix (pass) and 2.48×10^{-1} before (fail). Against the independent reference, the pre-fix absolute field is $1/h^3$ too large and the pre-fix worst per-component relative error rises toward 1 as h is refined: at $\gamma = 1$, aspect 1 it is 6.90×10^{-3} at $h = 1$, 6.78×10^{-1} at $h = 1/2$, and 9.63×10^{-1} at $h = 1/4$, while the after-fix values are stable to about 9×10^{-4} over $h = 1$ to $1/2$. (The aspect 4 series is 2.98×10^{-2} at $h = 1$, 5.98×10^{-1} at $h = 1/2$, 9.41×10^{-1} at $h = 1/4$.) At $h = 1$ the pre- and post-fix validate files are byte-identical, so the fix changes nothing where the original was already correct.

6.7 Determinism and the `gridUnitSI` defect

Repeated runs of the shipped CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes) [worker, 2026f]. The CLI also rejects a below-floor `-tolerance` (10^{-15}) and a two-velocity input, each with exit 1 [worker, 2026f].

A defect found during this validation and fixed after v0.1: `gridUnitSI` was applied twice. The reader converts the raw `gridSpacing` to SI (`OpenPMD.cpp:505`), and the CLI (`main.cpp:248`) multiplies by `gridUnitSI` a second time, so the cell size is off by one factor of `gridUnitSI`. A mesh with `gridUnitSI` $= 10^{-6}$ and a physical cell of 10^{-6} m yields a cell of 10^{-12} m and then aborts with `cannot create std::vector larger than max_size()`. Fixed in v0.1.1 with an end-to-end test that fails on the old code [worker, 2026f]. This is a lesson about reader-only unit tests: the reader-level conversion is correct, but the composition at the CLI boundary is what the user sees, and it was wrong.

Table 5: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: not reached) [worker, 2026c]. T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger. The $\gamma = 1$ aspect 1 row is $d = 8$ against both references here, versus the $d = 12$ of the shipped-CLI series in Table 4: the two measurements use different component sets (six axis directions, L2-pooled, versus the per-component relative error over x, y) and different implementations (the N2b standalone measurement versus the shipped executable), so they are not the same crossover [worker, 2026c,f].

	3×10^{-4}		10^{-3}			
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

7 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Neither is a truth comparison.

7.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 6), produced by the same header-only kernel PIConGPU calls with `fieldInitializer=referenceOnly` [worker, 2026g]. The measured binary is `339ca7ad`; its field path is unchanged through the shipped v0.1.3 (`da3cd423`), and `ctest` passes 10/10 at that measured commit [worker, 2026h]. Both sides use identical Yee offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation [worker, 2026h].

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B) [worker, 2026h, Orchestrator, 2026e]. picBunch is about four to six orders of magnitude above that floor. This is a no-agreement statement at the reference floor, not a defect: B is exactly zero for $\gamma = 1$ in both, the bulk agrees to below 1–3%, and the error is concentrated at the core and the beamed peaks [worker, 2026h, Orchestrator, 2026e].

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8 $\times 10^{-3}$ with an L2 of 4.6–5.7 $\times 10^{-2}$; $\gamma = 10$ bulk median 1.4–3.3 $\times 10^{-2}$ but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92 [worker, 2026h]. The forward component E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse components E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake that TSC and the spectral near-field kernel cannot resolve [worker, 2026h].

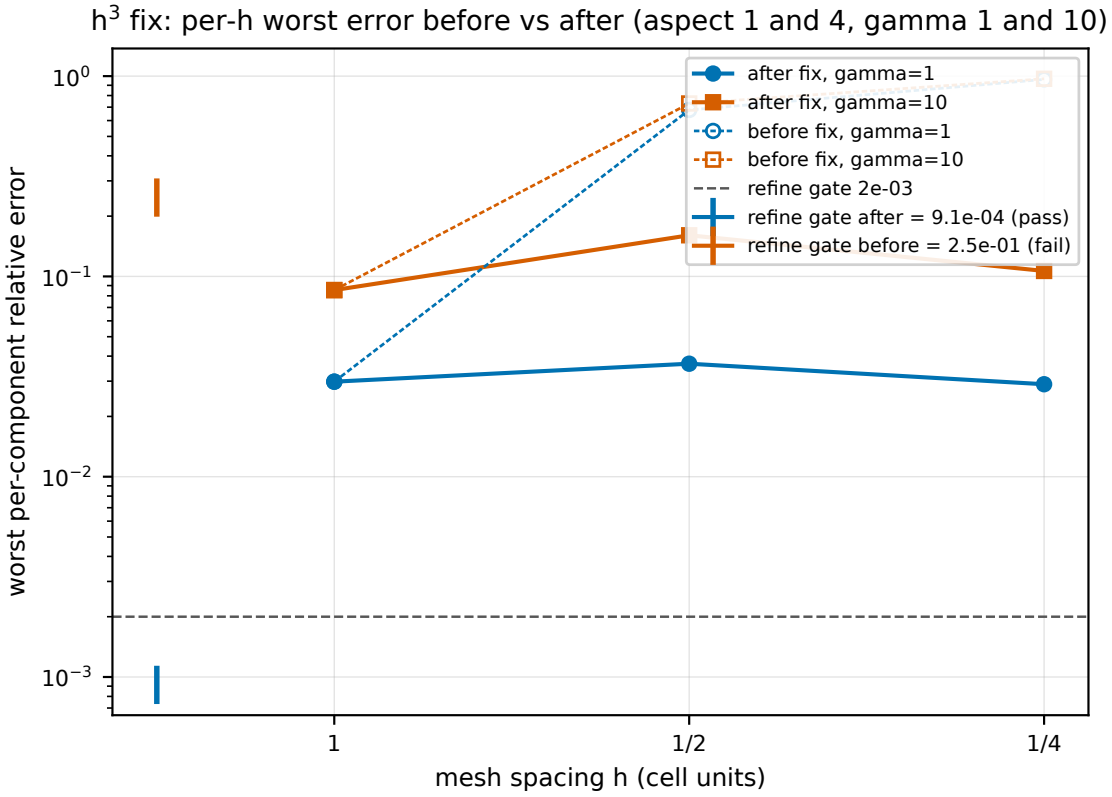


Figure 5: Measured per-h worst per-component relative error before and after the h^3 fix (commit `fef43bbd`, branch `experimental/mesh-h3`), at fixed physical separation, $\gamma \in \{1, 10\}$, aspect 1 and 4, rosi_v100 [worker, 2026a]. Solid markers, after the fix (stable to about 9×10^{-4} over $h = 1$ to 1/2); open markers, before the fix (relative error rises toward 1 as h is refined). The vertical bars are the refine-gate values $|\mathbf{h}{=}1 - \mathbf{h}{=}1/2|/(1 + \max)$: $9.13{\cdot}10^{-4}$ after (gate $2{\cdot}10^{-3}$, pass) and $2.48{\cdot}10^{-1}$ before (fail). At $h = 1$ the pre- and post-fix fields are byte-identical (sha256 `b6600ec0`...) [worker, 2026a].

Caveat: the sampling groups are distance-from-centroid. The 318-point groups (inside-support, face, outside) were cut by distance from the bunch centroid, in a bunch with a ~ 6-cell spread [Orchestrator, 2026e]. That grouping may understate how close the "outside" points actually are to a source's TSC support, so the per-group numbers are a spatial profile, not a clean inside/outside-of-support split. The per-source dependence in Section 6 uses the explicit support union and is the measurement that separates the per-source floor from any superposition contribution [worker, 2026g].

7.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0, E, B $x/y/z$ float32 at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata [worker, 2026i]. The remaining differences against a genuine PIConGPU checkpoint are non-blocking metadata [worker, 2026i].

Field arrival is bitwise. A validation-only build with the field solver set to `None` loads the checkpoint and writes step 1; all six components are bitwise-equal, max ULP 0 [worker, 2026i].

The shipped v0.1.3 defaults were also checked on the real LWFA checkpoint with only the restart-step and unit flags: picBunch took the $64 \times 128 \times 64$ extent and the origin 0 from the

Table 6: Whole-field comparison of picBunch against the exact-quadrature reference, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100 [worker, 2026h]. `worst_rel` is peak-normalized; median and p99 are per-point over nodes with |ref| $> 10^{-3}\times$peak; L2 is pooled. Not a truth comparison.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

mesh, and PIConGPU restarted the result with `-periodic 0 0 1`, exit 0, initialization time 4.303 s. In a validation-only build all six components arrived bitwise equal, max ULP 0 [worker, 2026j]. This is a solver=`None` round-trip: it checks transport, geometry, and units, not the field solver.

7.3 PML passthrough, the LWFA open-boundary limit, and the restart fixes

On a $24 \times 24 \times 12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the $\sim$ 116 s neuralBunch path to 1.8 s because the checkpoint carries the field [worker, 2026i]. The larger LWFA case is where the limit shows. A $64 \times 128 \times 64$ LWFA run with PML active produces a genuine PIConGPU checkpoint with 96 `Convolutional PML` E/B records. At the W7B commit picBunch does not write them, so a restart with open boundaries (`-periodic 0 0 1`) fails with `Key 'Convolutional PML E_xneg' does not exist (read-only)`, while `-periodic 1 1 1` on the same checkpoint restarts with exit 0, isolating the blocker to the missing PML records [worker, 2026i]. A later fix run (commit `bed4344c`, the PML-copy writer, W3f) closes this limit: it reads the real LWFA checkpoint with its 638,976 particles directly and copies the eight `Convolutional PML` groups, 48 datasets, float32, bitwise equal to the source, so the open-boundary restart (`-periodic 0 0 1`) then exits 0 with initialization 3.986 s [fix-run worker, 2026]. The PML state is required unconditionally by the checkpoint field list and loaded by the same loop, so a field-only writer without PML records cannot restart an open-boundary deck [worker, 2026i, fix-run worker, 2026].

The same investigation found the defects in PIConGPU's restart path that the fixes PR addresses [Orchestrator, 2026f]: `RestartFieldLoader` never reads `unitSI`, `gridSpacing`, or `gridGlobalOffset`; the PML slabs are unconditional; a successful restart suppresses the initial dump; and the openPMD backend does not convert dataset dtypes. The fixes PR (`psychocoderHPC-bot/picong` PR #25, draft) adds restart metadata validation, unitSI conversion, optional PML slabs, and an explicit dtype contract, verified by real restarts including a factor-2 unitSI case [Orchestrator, 2026f].

8 Discussion and limitations

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method.

What it is good for. In the far field the relative error against the point field is below 10^{-3} on the beam axis from $d = 6$ for every γ [worker, 2026c]. In relative per-component error against the TSC-convolved continuum: $\gamma = 1$, aspect 1 crosses 3×10^{-4} at $d = 12$ (1.1×10^{-4}); $\gamma = 1$, aspect 4 at $d = 32$ (2.6×10^{-4}); $\gamma = 10$ does not cross within 32 cells, flooring near 6×10^{-3} [worker, 2026f]. At 2×10^6 particles in a 20-cell box the compute-only total is 64.8 ms, dominated by the deposit (56.5 ms), and flat in γ [worker, 2026b]; end to end with the writer, 47.15 ms at 10^6 [worker, 2026f]. Repeated runs are bit-identical (md5 `d3641d64...`) [worker, 2026f]. The method is GPU-portable through the alpaka backend [worker, 2026b], and the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14} [worker, 2026e].

What it is not. The near shell is not implemented in v0.1: the delivered near-field tables were off by 1.9×10^{-2} to 3.3×10^{-1} against the 3×10^{-4} target on all six [Orchestrator, 2026b]. The perpendicular direction at $\gamma = 10$ does not converge at the natural cell size: 1.5×10^{-2} at $d = 48$, aspect 1, against the point field, of which the TSC smoothing alone is 1.48×10^{-2} (Table 3). Reaching 10^{-3} perpendicular at $r = 8$ cells would need $h \approx 1/29$, a 2.4×10^4-fold increase in M^3 FFT work [worker, 2026c, Orchestrator, 2026c]. At aspect 4, $\gamma = 1$ already fails the gate at $d = 48$ (2.4×10^{-3}) [worker, 2026c]. The cross-validation confirms the same boundary: the mesh does not agree with the exact reference at the reference floor, the disagreement is per-source, concentrated in the TSC support, and does not grow with particle count [worker, 2026g, Orchestrator, 2026e].

Performance. The cost is two regimes with a crossover set by the box, not the particle count [worker, 2026b]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows. At large boxes the FFT and kernel dominate: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× [worker, 2026b]. The timings are single-GPU on one `rosi_v100` host; there is no multi-GPU or multi-rank measurement [worker, 2026b].

Scope of the accuracy statements. The defensible v0.1 scope, in relative per-component error against the continuum, is $\gamma = 1$ aspect 1 from $d = 12$, $\gamma = 1$ aspect 4 from $d = 32$, and the axial component at $\gamma \leq 10$ [worker, 2026f,c, Orchestrator, 2026c]. The blanket statement "$\gamma \leq 10$ meets 10^{-3}" is not supported: it fails perpendicular at the natural cell size and floors near 6×10^{-3} at $\gamma = 10$, $d = 32$, aspect 1 [worker, 2026f,c]. The judge metric $|\text{ref} - \text{mesh}|/(1 + \max)$ is absolute-dominated for fields of order 10^{-9} V/m and reports $d = 2$ for every case; it does not discriminate. The relative per-component error is the metric of record [worker, 2026f]. The radii in Table 5 are axis directions only; oblique directions are $\sim 1.5\times$ larger [worker, 2026c]. Scale invariance (Section 3.5) means only r/h moves the error: at a fixed physical distance, refining h raises r/h and does reduce the perpendicular error, but at the M^3 cost in $1/h$, so $h \approx 1/29$ (2.4×10^4 times the M^3 nodes) to reach 10^{-3} at $r = 8$ [worker, 2026c, Orchestrator, 2026a].

9 Conclusion

picBunch computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, and the discrete recipe, its measured cost, and its accuracy are given in full. In relative per-component error against its TSC-convolved target, the mesh is below 3×10^{-4} from $d = 12$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$ it floors near 6×10^{-3} at $d = 32$ and does not cross within the swept range [worker, 2026f]. The disagreement with the exact quadrature reference is per-source, local to the TSC support, and independent of particle count. On periodic decks the field round-trips bitwise through a real PIConGPU checkpoint and restart: a solver=None validation load writes step 1 and all six components are bitwise-equal, max ULP 0. The method is a far-field tool: cheap, deterministic (md5 `d3641d64...`, bit-identical repeats), and GPU-portable,

with the near shell absent in v0.1 and the perpendicular direction at $\gamma \geq 10$ not converged at the natural cell size.

References

S1 extraction worker. picbunch portable algorithm specification (s1), 2026. Run pr24-20261102. S1-algorithm-spec.md. Exact field definition and discrete recipe with path:line provenance. meshconv.cu on branch fee89ed75; ExactShapeField.hpp on 2baa2566.

F4 worker. The h^3 scaling defect, before and after the fix (f4), 2026a. Run pr24-20261102. F4/ directory. Per-h series in f4_per_spacing_after.csv and f4_per_spacing_before.csv, worst per-component relative error against indfield.py at fixed physical separation, aspect 1 and 4, gamma 1, 10, 100, rosi_v100. Refine gate: after 9.13e-4 (pass), before 2.48e-1 (fail). At h=1 the pre- and post-fix validate files are byte-identical (sha256 b6600ec0...). See F4/COMMANDS.txt.

SB1 worker. Measured scaling of the picbunch field pipeline (sb1), 2026b. Run pr24-20261102. SB1/RESULTS.md. V100 timing matrix, Coulomb limit, distance sweep. Commit dc69c16a21f1dc330746365e3409560cb2399ed9.

N2b worker. picbunch mesh path vs the tsc-convolved continuum and the point field (n2b), 2026c. Run pr24-20261102. N2b/RESULTS.md. Reference-separated accuracy, scale invariance, crossover table. Arbiter: indfield.py sha256 a135598041a3dbe134667585631ea53fc6bdb7cf7a84419b5a8d762202599519.

N1b worker. picbunch mesh convolution measurements, cpu (n1b), 2026d. Run pr24-20261102. N1b/RESULTS.md. Wraparound control, R2C/C2R vs C2C, padding vs distance, kernel sampling rule, CPU cost model. CPU only, no GPU.

Orchestrator. Decision d10: mesh path accuracy is scale-invariant in h at fixed distance-in-cells, 2026a. Run pr24-20261102. DECISIONS.md, D10. Kernel homogeneous of degree -2, TSC width equals node spacing. Measured identical errors at h = 1, 1/2, 1/4, 1/8.

SB1 worker. Coulomb-limit control (sb1), 2026e. SB1/RESULTS.md and PF/data/coulomb.csv. u = 0, one particle at a node, far nodes at d = 8, 16, 24, 32, six axis directions. Worst relative error 4.88e-14 (max absolute 2.28e-25). Threshold 1e-12: PASS.

W4b worker. Independent validation of picbunch at the v0.1 pipeline tip (w4/w4b), 2026f. Run pr24-20261102. W4b/RESULTS.md (mirror of W4/). picBunch at bfaa1e26 (experimental/mvp), rosi_v100. ctest 8/8. Accuracy: shipped CLI vs indfield.py order 12 refine 3, relative per-component error per d = 2 to 32 cells, gamma 1 and 10, aspect 1 and 4, single and 3-particle cloud. Relative 3e-4 crossover: d=12 (gamma 1 aspect 1), d=32 (gamma 1 aspect 4), none within 32 for gamma 10 (plateau 6e-3). Judge metric is absolute-dominated: d=2 for every case. End-to-end timings with writer (20-cell box, gamma 10, 3 repeats): 1e4 total 22.00 ms; 1e6 total 47.15 ms. Determinism: md5 d3641d64, 15092736 bytes, bit-identical. gridUnitSI applied twice: 1e-6 grid gives 1e-12 m cell, aborts with max_size exception. Fixed in v0.1.1 with an end-to-end test.

Orchestrator. Decision d4: near-table route closed, 2026b. Run pr24-20261102. DECISIONS.md, D4. First real measurement: 1.9e-2 to 3.3e-1 on all six tables; does not meet the 3e-4 target.

Orchestrator. Decision d11: v0.1 accuracy scope, 2026c. Run pr24-20261102. DECISIONS.md, D11. Not defensible: gamma <= 10 meets 1e-3. Defensible: gamma <= 1 aspect 1; gamma <= 10 axial only; gamma 100 axial only.

Orchestrator. Decision d8: gamma-100 outside-band 9.43e-2 is unresolved, 2026d. Run pr24-20261102. DECISIONS.md, D8. Pre-existing 9.43e-2 at 1.6 to 6 cells, gamma 100 axial, outside the near box.

pr24w53 ref. indfield.py: independent tsc-convolved continuum reference, 2026. sha256 a135598041a3dbe134667585631ea53fc6bdb7cf7a84419b5a8d762202599519. Tensor Gauss-Legendre with TSC-knot split and singularity-cancelling power map. The independent truth against which the mesh is validated.

W7t worker. Reference-quantity identification and particle-count scaling (w7t), 2026g. Run pr24-20261102. W7t/RESULTS.md (mirror of /home/ubuntu/pr24/w7s-20261102/RESULTS.md). The exact-quadrature reference is the TSC-convolved field, the same quantity picBunch computes (exact_field_driver.cpp:123,127; ReferenceKernel.hpp:625,131,142,281,428,522,596; ExactShapeField.hpp:589,157,222,472). N-count scaling: disagreement is per-source, not superposition; B identically zero for the stationary bunch.

W7r worker. picbunch cross-validation against picongpu's exact field reference (w7/w7r), 2026h. Run pr24-20261102. W7r/RESULTS.md (re-verification of W7/RESULTS.md). picBunch @ 339ca7ad3903249ebf3859f5e106d63db4dc3df5. 318 staggered points (64x64x32, cell 1e-7 m) and whole-field (N=1000, 32x32x16, cell 2e-7 m). Groups are distance-from-centroid (caveat D13). picBunch does not agree with the reference at the reference floor (3.66e-2 gamma 1; 5.16e-1 gamma 10 E/y vs floor 1.19e-6 E / 1.43e-6 B).

Orchestrator. Decision d13: cross-validation against picongpu does not show agreement at the reference floor, 2026e. Run pr24-20261102. DECISIONS.md, D13. 3.66e-2 (gamma 1), 5.16e-1 (gamma 10 E/y) vs floor 1.19e-6 (E), 1.43e-6 (B). No claim that picBunch agrees with PIConGPU.

W7b worker. picbunch checkpoint acceptance on picongpu pr 24 (w7b), 2026i. Run pr24-20261102. W7B/RESULTS.md. Early acceptance state: 16-particle periodic deck restarts with exit 0 and all six components bitwise (0 ulp) through a solver=None validation load; initialization 1.8 s vs the 116 s neuralBunch path. The larger LWFA deck (64x128x64, 96 Convolutional PML records) did not restart with open boundaries at this commit because picBunch did not yet write the PML records (Key 'Convolutional PML E xneg' does not exist); periodic restart of the same checkpoint exited 0. picBunch fix commit 0c9427dee82e01c832c42a8358033b701386f37e.

PW worker. picbunch v0.1.3 defaults-only lwfa checkpoint acceptance, 2026j. Run pr24-20261102. Defaults-only restart on the real LWFA checkpoint at tag v0.1.3 (da3cd4239926f5da31b5794a36f8184e0b3164da), only restart-step and unit flags. 64x128x64 grid and origin 0 from the mesh. PIConGPU –periodic 0 0 1, exit 0, initialization 4.303 s. Validation-only solver=None: all six components bitwise, max ULP 0.

W7 fix-run worker. picbunch checkpoint restart with pml passthrough (fix run), 2026. Run pr24-20261102. Later fix run on branch experimental/mvp at bed4344c45c46a5ec77529b652631d75f7d5bb5b (parent of the v0.1 tag e702002), with the PML-copy writer (W3f). Recorded in docs/picongpu-restart.md of the W3f tree. 638,976 particles read directly from the real LWFA checkpoint (w7b-20261102/lwfa/gt-run/checkpoints/out 000002 h5); the writer copies the eight Convolutional PML groups, 48 datasets, float32, bitwise equal to the source. Open-boundary restart (–periodic 0 0 1) exits 0, initialization 3.986 s. Periodic case: all six components bitwise, 0 ulp.

Orchestrator. Decision d15: Picongpu fixes pr #25 (restart path), 2026f. Run pr24-20261102. DECISIONS.md, D15. psychocoderHPC-bot/picongpu PR #25 (draft) adds restart metadata validation, unitSI conversion, optional PML slabs, and a dtype contract.